Machine Learning - Undergraduate Level Assessment 15 Total Marks: 40 Answer all questions.

Multiple Choice (10 marks)

1. Which of the following are the spatial clustering algorithms?
   (a) Partitioning based clustering
   (b) K-means clustering
   (c) Grid based clustering
   (d) All of the above
2. Statement 1| Overfitting is more likely when the set of training data is small. Statement 2| Overfitting is more likely when the hypothesis space is small.
   (a) True, True
   (b) False, False
   (c) True, False
   (d) False, True
3. Statement 1| RoBERTa pretrains on a corpus that is approximate 10 x larger than the corpus BERT pretrained on. Statement 2| ResNeXts in 2018 usually used tanh activation functions.
   (a) True, True
   (b) False, False
   (c) True, False
   (d) False, True
4. K-fold cross-validation is
   (a) linear in K
   (b) quadratic in K
   (c) cubic in K
   (d) exponential in K
5. The numerical output of a sigmoid node in a neural network:
   (a) Is unbounded, encompassing all real numbers.
   (b) Is unbounded, encompassing all integers.
   (c) Is bounded between 0 and 1.
   (d) Is bounded between -1 and 1.

True / False (10 marks)

Answer True or False

6. True or False: The polynomial degree significantly impacts the trade-off between underfitting and overfitting in polynomial regression.
7. True or False: The cost of one gradient descent update, given the gradient, is O(D), where D is the number of dimensions in the parameter space.
8. True or False: Applying an L_2 norm regularization penalty will lead to some coefficients being exactly zero in linear regression.
9. True or False: Bayesians advocate for the use of prior distributions on parameters in probabilistic models, while frequentists do not.
10. True or False: High entropy indicates that the partitions in classification are not pure, reflecting a mix of different classes.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Write a function to sort a list of tuples using lambda.
12. Write a function to search some literals strings in a string by using regex.

End of Paper